

$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \dots$$

5 6 3

- b) Use D'Alembert's Ratio Test to find the interval of convergence of the following series

5 6 3

$$1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

- c) Test the convergence of the following series using Cauchy's Root Test:

6 4 4

$$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n$$

- Q5. a) Prove that the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is conditionally convergent.

5 6 3

- b) Determine the interval of convergence for the following power series

5 6 3

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

- c) Obtain the Maclaurin series for the function $f(x) = \sin(x)$.

6 6 3



DAYANANDA SAGAR
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SCHOOL OF
ENGINEERING

USN No: ENG25CS0453

B. Tech II Semester End Examinations – May 2026

Course Title: C Programming for Problem Solving

Course Code: 25EN1114

Duration: 02 Hours 30 Minutes

Date: 25-05-2026

Time: 01:00 PM to 03:30 PM

Max Marks: 80

- Note:**
1. Answer all Five Full Questions
 2. Each Full Question carries 16 Marks
 3. Draw neat sketches wherever necessary
 4. Missing Data may be suitably assumed

Q. No.	Question	Marks	BTL	COs
1.a.	Design an algorithm and flowchart to find biggest among three numbers.	05	L2	CO1
1.b.	Predict the output for the following code: <pre>#include <stdio.h> int main() { int a = 5, b = 10, c; c = a++ * --b + b / a; printf("%d %d %d", a, b, c); return 0; }</pre> 3, 9, 46 <pre>#include <stdio.h> int main() { int a = 5, b = 9, c = -8, d = 1, z; z = a && b > 10 && c d printf("%d", z); return 0; }</pre> 1	06	L2	CO1
1.c.	Write a C program to read the <u>name of the user</u> , number of <u>units</u> consumed and print out the charges. An electricity board charges the following rates for the use of electricity: for the first <u>100</u> units 80 paise per unit: for the <u>next</u> 100 units 90 paise per unit: beyond 300 units Rs 1 per unit. If the total amount is more than <u>Rs 400</u> , then an additional surcharge of 15% of total amount is charged.	05	L3	CO1
2.a.	Demonstrate selection sort on an array of 5 integers [64, 25, 12, 22, 11] with intermediate steps.	05	L3	CO2

(P.T.O.)